Operation Management

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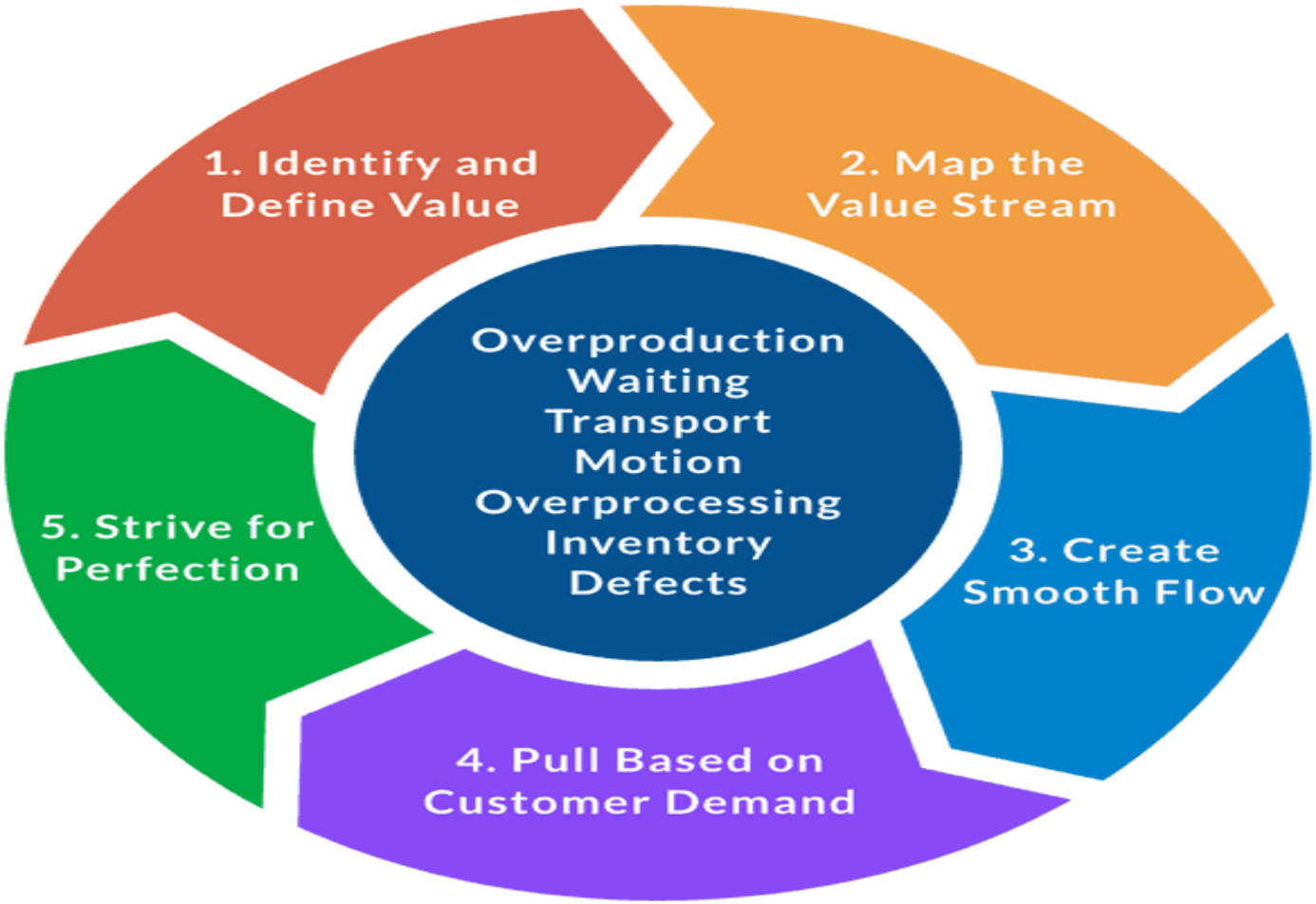
Lecture No. 14

Lean Operations

- Lean is a systematic approach to identify and eliminate nonvalue added activity through continuous improvement by flow of the product at the pull of the customer in pursuit of perfection.
- It involves everyone in the organization for the minimization or elimination of non-value added activities and to maximize customer value in lesser resources.

Principles of Lean



Source: Lean Production

Identify Value

Value : Something for which customers are willing to pay for.

Value added Activity (VA): Any activity that changes the form, fit or function of materials or information to meet customer's expectations and the customer is willing to pay for it.

Non- Value-added Activity (NVA): Any activity that changes the form, fit or function of materials or information to meet customer's expectations but the customer is not willing to pay for it.

Identify Value

- Necessary Non-Value-added Activity (NNVA): Any activity for which a customer is not willing to pay, but the activity is necessary for the manufacturer.

Map the Value Stream

- Value Stream: Set of all the specific actions required to bring a specific product produced from raw materials to the customer.
- O = Optimize Value added activities.
- E= Eliminate Non-value added activity.
- M= Minimize NNVA.

7 + 3 types Waste

- T = Transportation
- I = Inventory
- M= Motion
- W = Waiting
- O = Over Production
- O = Over Processing
- D = Defect

1. Under Utilized Skills.
2. Wrong Use of Automation.
3. Wrong Use of Metrics.

Create Flow

- Right things to do in the right sequence at the right place.
- Focus on specific design, order, features, or process.
- Ignore traditional boundaries of facts or departments.

Establish Pull

- Make to Order
- JIT
- JIT is a philosophy which eliminates waste associated with time, labour, and storage space.
- Company will produce what is needed or what the customer wants.

Establish Pull

Figure 16.1

Lean Contributes to Competitive Advantage

JIT TECHNIQUES:

Suppliers:	Few vendors; supportive supplier relationships; quality deliveries on time, directly to work areas
Layout:	Work cells; group technology; flexible machinery; organized workplace; reduced space for inventory
Inventory:	Small lot sizes; low setup time; specialized parts bins
Scheduling:	Zero deviation from schedules; level schedules; suppliers informed of schedules; kanban techniques
Preventive maintenance:	Scheduled; daily routine; operator involvement
Quality production:	Statistical process control; quality suppliers; quality within the firm
Employee empowerment:	Empowered and cross-trained employees; training support; few job classifications to ensure flexibility of employees
Commitment:	Support of management, employees, and suppliers

WHICH RESULTS IN:

- Rapid throughput frees assets
- Quality improvement reduces waste
- Cost reduction adds pricing flexibility
- Variability reduction
- Rework reduction

WHICH WINS ORDERS BY:

Faster response to the customer at lower cost and higher quality

A Competitive Advantage

Source: Heizer, Render, Muckson

Seek Perfection

- We should reassess periodically the process for perfection.

Source: Internet

Result

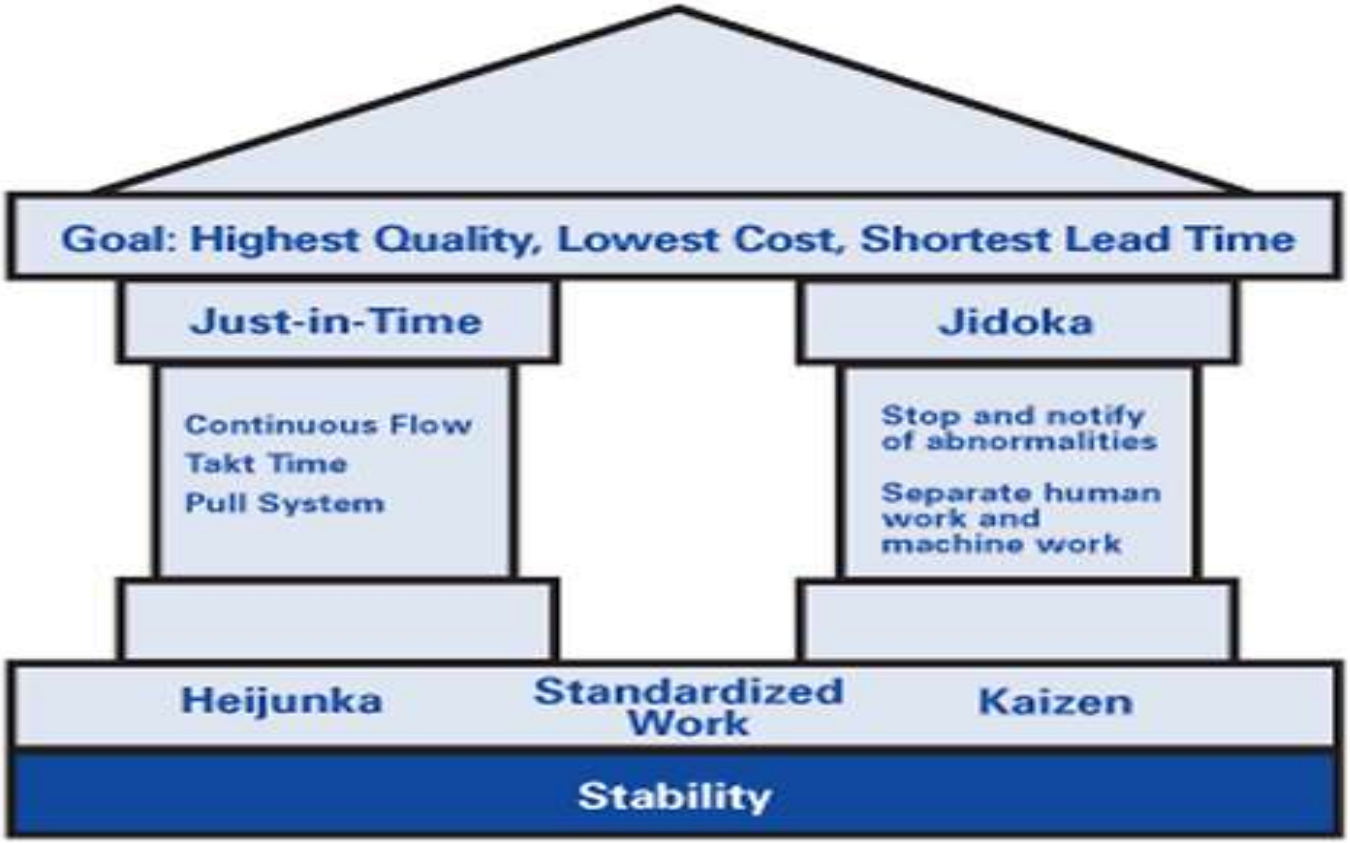
- Eliminate Waste.
- Remove Variability.
- Improve Throughput.
- Reduces Cycle Time.

Any Question

TPS

- Toyota Production System
- Goal : Highest Quality, Lowest Cost, Shortest Lead Time.
- Pillars of TPS : JIT, Jidoka, Heijunka, Standardized Work, Kaizen and Respect for people.

TPS

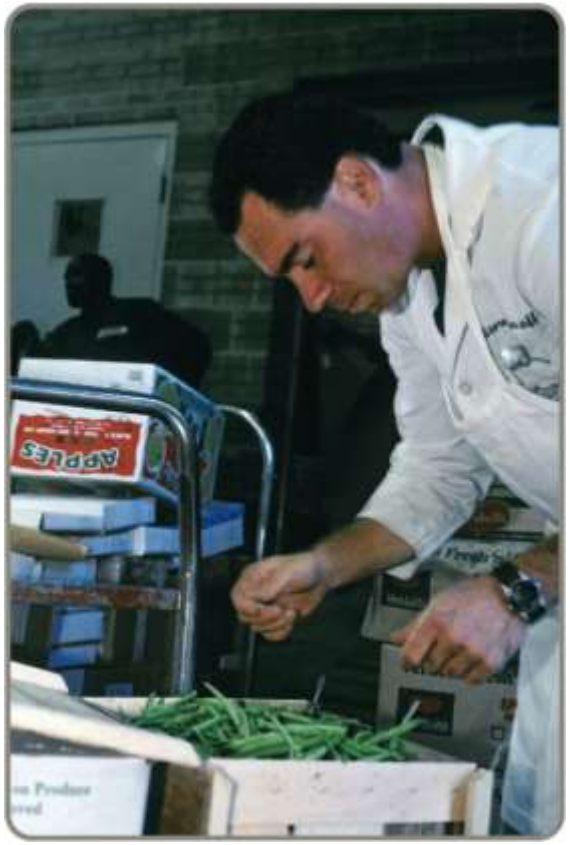


Toyota Production System "House."

Source: Internet

Just-In-Time

- Powerful strategy for improving operations
- Materials arrive *where* they are needed only *when* they are needed
- Identifying problems and driving out waste reduces costs and variability and improves throughput
- Requires a meaningful buyer-supplier relationship



Jidoka

- Highlights the causes of the problem.
- Autonomation (Automation + Human intelligence).
- Here machine detects a problem and communicates through display board.
- It separate human work and machine work.

Heijunka

- Japanese word of levellization.
- Level the work load.

Standardized Work

- Work is completely specified as to content, sequence, timing, and outcome; this is fundamental to a good process.
- Supplier connections for both internal and external customers are direct, specifying personnel, methods, timing, and quantity.
- Material and service flows are simple and directed to a specific person or machine.

Kaizen

- Kai + Zen.
- Small changes + Improvement.
- Continuous improvement in small increments.

Respect for People

- Employees are empowered.
- They are empowered not only to make improvements, but also to stop machines and processes when quality problems exist.

Any Question

Lean Organization

1. Product design.
2. Process design.
3. Personnel/organizational elements.
4. Manufacturing planning and control.

Product Design

1. Standard parts.
2. Modular design.
3. Highly capable production systems with quality built in.
4. Concurrent engineering.

Process Design

- Lean Layout.
- Lean Inventory.
- Lean Scheduling.

▸ Reduce waste due to movement

Lean Layout

TABLE 16.1
LEAN LAYOUT TACTICS
Build work cells for families of products
Include a large number operations in a small area
Minimize distance
Design little space for inventory
Improve employee communication
Use *poka-yoke devices*
Build flexible or movable equipment
Cross-train workers to add flexibility

Distance Reduction

- Large lots and long production lines with single-purpose machinery are being replaced by smaller flexible cells
- Often U-shaped for shorter paths and improved communication
- Often using group technology concepts

Increased Flexibility

- Cells designed to be rearranged as volume or designs change
- Applicable in office environments as well as production settings
- Facilitates both product and process improvement

Impact on Employees

- Improved communications facilitate the passing on of important information about the process.
- With little or no inventory buffer, getting it right the first time is critical

Lean Inventory

- Use a pull system to move inventory.
- Reduce lot size
- Develop just-in-time delivery systems with suppliers.
- Deliver directly to the point of use.
- Perform to schedule.
- Reduce setup time.
- Use group technology.

Reduce Lot Size

- Take Production Order Quantity Model.

$$Q_p^* = \sqrt{\frac{2DS}{H[1-(d/p)]}} \qquad (16\text{-}1)$$

where D = Annual demand $\quad d$ = Daily demand

S = Setup cost $\quad p$ = Daily production

H = Holding cost

Source: Heizer, Render, Muckson

Class Exercise

- Crate Furniture, Inc., a firm that produces rustic furniture, desires to move toward a reduced lot size.
- Crate Furniture's production analyst, Aleda Roth, determined that a 2-hour production cycle would be acceptable between two departments.
- Further, she concluded that a setup time that would accommodate the 2-hour cycle time should be achieved

Class Exercise

D = Annual demand = 400,000 units

d = Daily demand = 400,000 per 250 days = 1,600 units per day

p = Daily production rate = 4,000 units per day

Q_p = EOQ desired = 400 (which is the 2-hour demand; that is, 1,600 per day per four 2-hour periods)

H = Holding cost = \$20 per unit per year

S = Setup cost (to be determined)

Hourly labor rate = \$30.00

Source: Heizer, Render, Muckson

Solution

SOLUTION ▶ Roth determines that the cost and related time per setup should be:

$$Q_p = \sqrt{\frac{2DS}{H(1 - d/p)}}$$

$$Q_p^2 = \frac{2DS}{H(1 - d/p)}$$

$$S = \frac{(Q_p^2)(H)(1 - d/p)}{2D} \qquad (16\text{-}2)$$

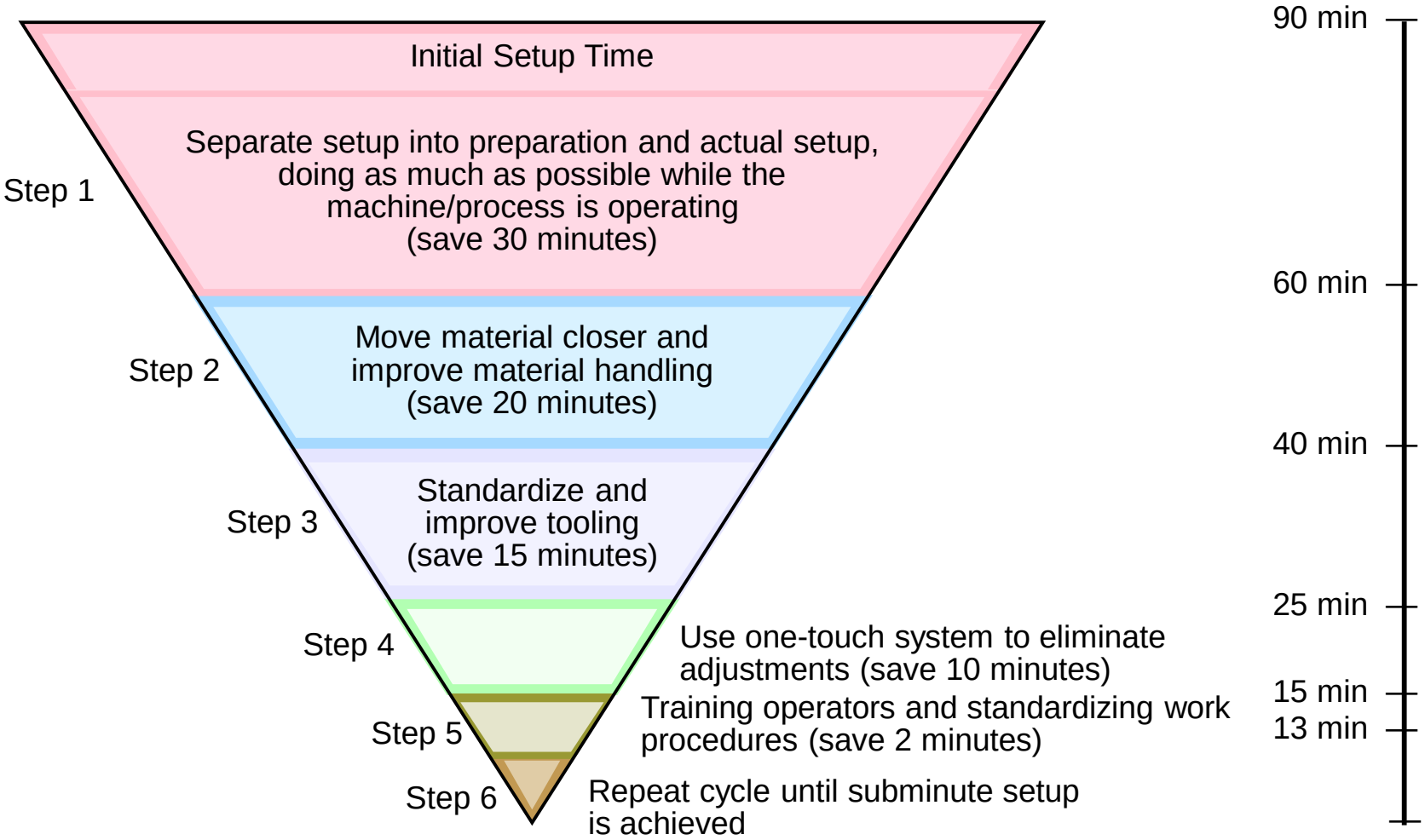
$$= \frac{(400)^2(20)(1 - 1{,}600/4{,}000)}{2(400{,}000)} = \frac{(3{,}200{,}000)(0.6)}{800{,}000} = \$2.40$$

$$\begin{aligned}\text{Setup time} &= \$2.40/(\text{hourly labor rate})\\ &= \$2.40/(\$30 \text{ per hour})\\ &= 0.08 \text{ hour, or } 4.8 \text{ minutes}\end{aligned}$$

Source: Heizer, Render, Muckson

Reduce Setup Costs

Figure 16.6



Lean Scheduling

- **Better scheduling improves performance**

TABLE 16.3

LEAN SCHEDULING TACTICS
Make level schedules
Use kanbans
Communicate schedules to suppliers
Freeze part of the schedule
Perform to schedule
Seek one-piece-make and one-piece-move
Eliminate waste
Produce in small lots
Make each operation produce a perfect part

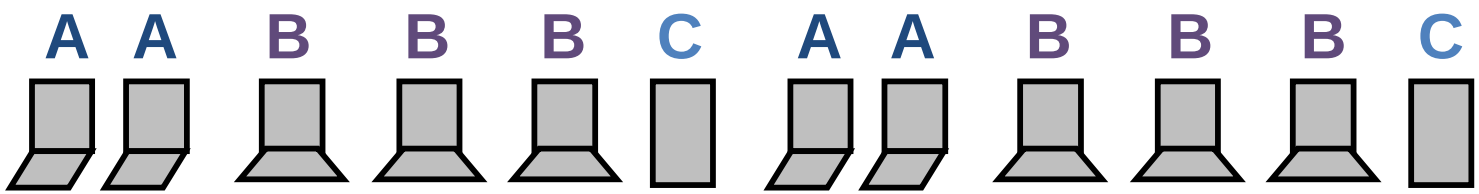
Level Schedules

- Process frequent small batches rather than a few large batches
- Make and move small lots so the level schedule is economical
- Freezing the schedule closest to the due dates can improve performance

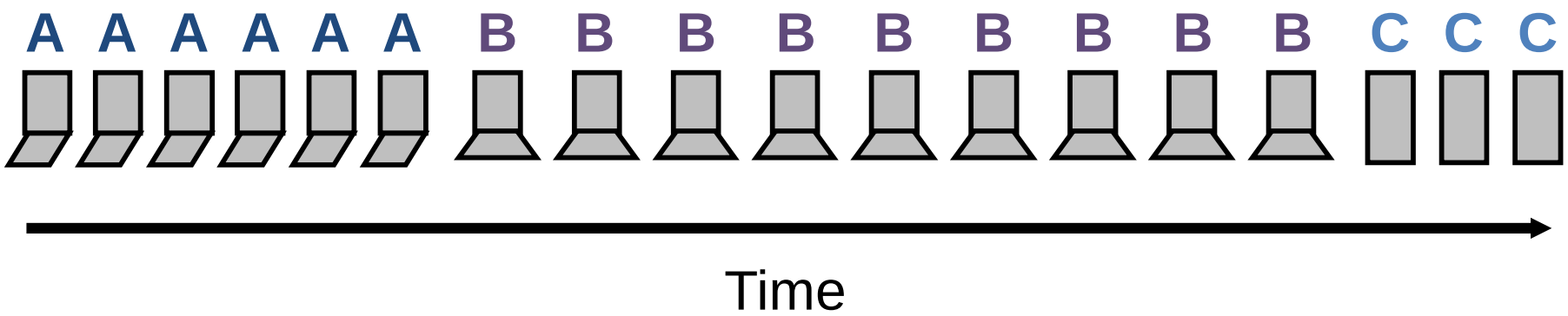
Scheduling Small Lots

Figure 16.7

JIT Level Material-Use Approach



Large-Lot Approach



Poka Yoke (Mistake Proofing)

- Introduced by Dr. Shigeo Shingo.
- Improvement strategy for mistake proofing to prevent defects from arising during production process.
- Error free design or process.
- Zero defect.
- Human error is always a factor.
- Can be achieved by more training, employee commitment and discipline.

SMED

- Single minute Exchange of Die
- A system for dramatically reducing the time to complete equipment changeover.
- Streamlining of complex changeover steps.
- It lowers manufacturing cost.
- Lowers inventory.
- Improved responsiveness.

Kanban

- The Japanese word for *card* , which has come to mean "signal"; a kanban system moves parts through production via a "pull" from a signal.

$$\text{Number of kanbans (containers)} = \frac{\text{Demand during lead time} + \text{Safety stock}}{\text{Size of container}} \qquad (16\text{-}3)$$

Source: Heizer, Render, Muckson

Class Exercise

- Hobbs Bakery produces short runs of cakes that are shipped to grocery stores.
- The owner, Ken Hobbs, wants to try to reduce inventory by changing to a kanban system.
- He has developed the following data and asked you to finish the project.

Class Exercise

- Production lead time = Wait time + Material handling time + Processing time = 2 days
- Daily demand = 500 cakes
- Safety stock = 1/2 day
- Container size (determined on a production order size EOQ basis) = 250 cakes

Solution

SOLUTION ▶ Demand during lead time =

$$\text{Lead time} \times \text{Daily demand} = 2 \text{ days} \times 500 \text{ cakes} = 1{,}000$$

$$\text{Safety stock} = \tfrac{1}{2} \times \text{Daily demand} = 250$$

Number of kanbans (containers) needed =

$$\frac{\text{Demand during lead time} + \text{Safety stock}}{\text{Container size}} = \frac{1{,}000 + 250}{250} = 5$$

Source: Heizer, Render, Muckson

Personnel / Organizational elements.

- Workers as assets.
- Cross-trained workers.
- Continuous improvement.
- Cost accounting.
- Leadership/project management.

Manufacturing Planning and Control

- Level loading.
- Pull systems.
- Visual systems.
- Limited work-in-process (WIP).
- Close vendor relationships.
- Reduced transaction processing.
- Preventive maintenance and housekeeping.

Supplier Partnerships

- **Supplier partnerships exist when a supplier and purchaser work together to remove waste and drive down costs**
- **Four goals of supplier partnerships are:**
  - *Removal of unnecessary activities*
  - *Removal of in-plant inventory*
  - *Removal of in-transit inventory*
  - *Improved quality and reliability*

JIT Partnerships

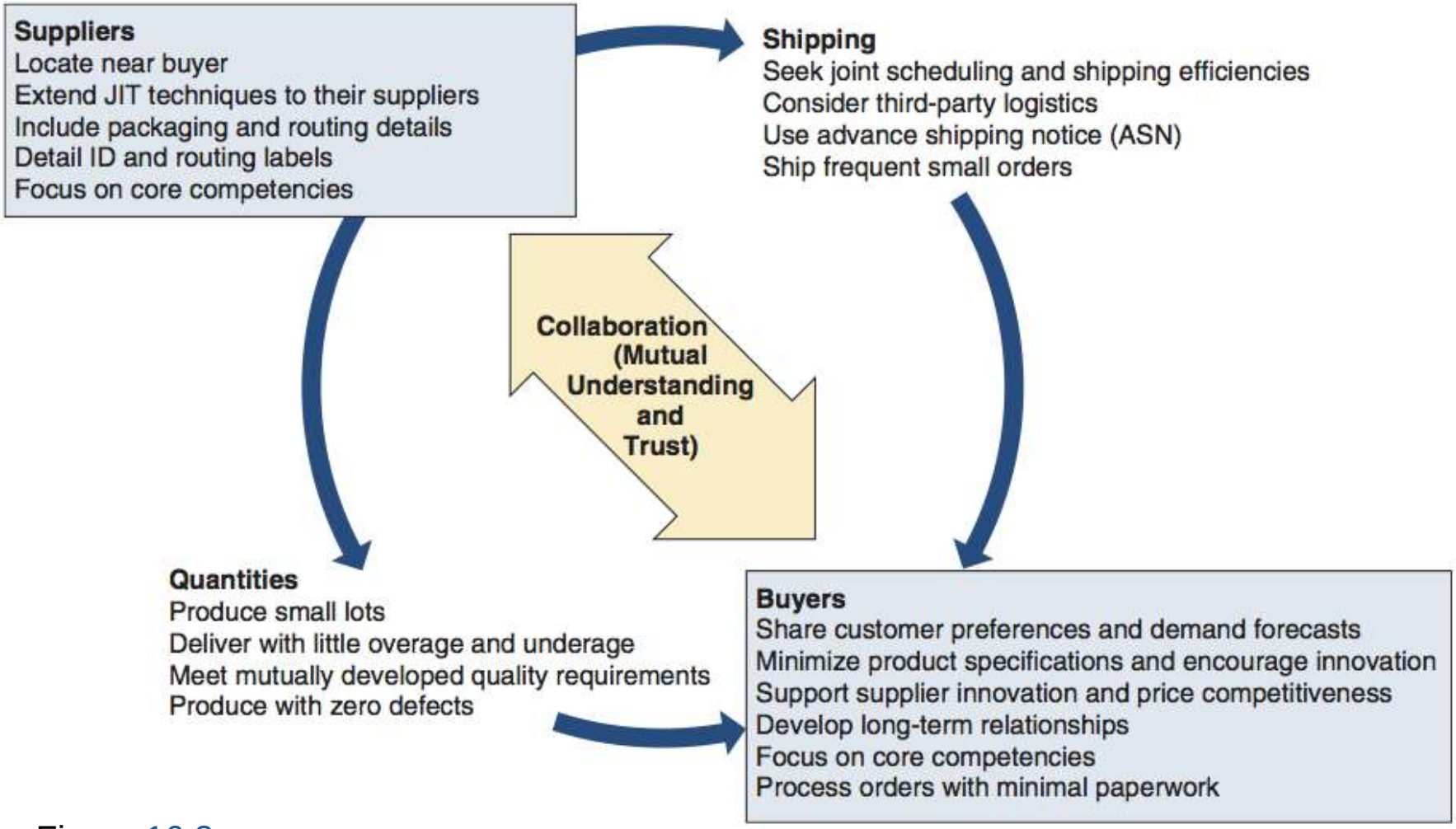


Figure 16.2

Lean in Services

- The features of Lean operations apply to services.
- Forecasts in services may be very elaborate, with seasonal, daily, hourly, or even shorter components.
- Lean is helpful to eliminate disruption, make the system flexible, eliminate the non-value added activities, minimize work in process and simplify the process.

Any Question

Further Reading / References

1. Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 16
2. Operation Management by William J Stevenson—Chapter 14
3. KPMG Handbook of Lean Management.
4. Lean Management—ASQ.

THANK YOU